



The best way to watch the MCU

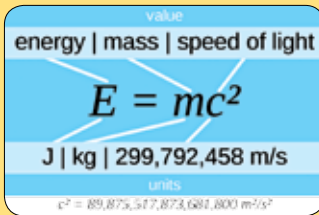
Meet the avengers, watch them assemble, and take a trip through space in this viewing order <https://dgit.in/may19-81>



The math of Skittles

How many packets of Skittles do you have to go through before finding two identical ones? <https://dgit.in/may19-82>

GEEK-KEY



MASS-ENERGY EQUIVALENCE

The mass energy equivalence, which states that energy is equal to mass multiplied by the speed of light squared, is called

the most famous equation in science. So famous that we dedicated our very first dmystify to explaining it! The equation ushered in a new era of physics, as it showed that energy and mass are different forms of the same thing, and can be turned from one form to another. The equation implies that there is a tremendous amount of energy packed into the stuff that we see all around us. Mass energy equivalence shows that it is possible to convert a tremendous amount of energy to very little mass, or very little mass can be converted to a tremendous amount of energy. The latter formed the theoretical basis for the development of the atom bomb. A paper clip, for example, could convert into an energy output equal to the yield of the bomb that was dropped on Hiroshima in 1945!



SCHRÖDINGER'S EQUATION

Erwin Schrödinger published the equation in 1926, which was an attempt to describe the behaviour of subatomic

particles. According to some interpretations, the equation is the most complete description of reality as we know it, and is applicable across all scales, from the tiniest subatomic particles, to the universe as a whole. One of the implications of the equation was that all particles, not just photons also behave like waves. In fact, not just particles, but everything, including this magazine, the objects in your environment, the Earth and our galaxy, all have a wavelength, but one so tiny that it is not noticeable. One problematic interpretation of the equation is that all possible realities are happening simultaneously. Given a choice between two outcomes with an equal chance of occurring, the universe itself splits into two, leading to parallel universes.

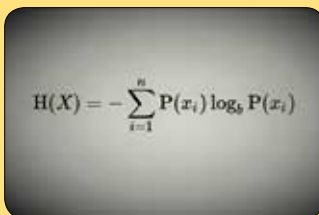


DIRAC EQUATION

In 1928, Paul Dirac derived an equation that described the behaviour of electrons at speeds close to that of light. The equation had some similarities to

Schrödinger's equation. Dirac did not himself fully appreciate the significance of his own equation. While the equation described the behaviour of electrons at relativistic speeds adequately, it had a caveat. It predicted the existence of a previ-

ously unknown particle, the positron, with the opposite charge as an electron. Now, this second particle could not be ignored, and the proof of the equation meant that the positron, or the antielectron had to exist. The equation predicted the existence of antimatter, which was then eventually discovered. The equation directly advanced the standard model, the current scientific understanding of how the universe works. The work of Dirac has assumed the same level of importance in Physics as Newton, Maxwell and Einstein. For their work in advancing the field of atomic physics, Schrödinger and Dirac shared the Nobel Prize for Physics in 1933.



SHANNON'S INFORMATION THEORY

This equation developed by Claude Shannon at Bell Labs during the advent of digital computing, forms

the basis of the modern information society. Given a number of letters and the length of the string, it is possible to quantify how much information is being transferred. However, languages use some letters more than others. For example, an e or an a is used much more frequently than a Q or a Z. Now, in the ASCII system, each letter of the alphabet is assigned a text string of equal length. A and E are represented by 01100001 and 01100101, while Q and Z are represented by 01010001 and 01011010. It is possible to reduce the footprint of a communication significantly by, say, representing frequently used A and E with 4 digits, and rarely used Q and Z with 12 digits. It is essentially the same idea that has given rise to compression formats such as ZIP, MP3 and JPEG.



THE DRAKE EQUATION

To initiate scientific dialogue on the first meeting of the search for extraterrestrial intelligence (SETI) project, physicist Frank

Drake wrote this equation in 1961. The equation is an attempt to calculate the number of intelligent extraterrestrial species in the Milky Way. The equation factored in the process of star formation, the number of systems with planets with conditions suitable for harbouring life, the length of time required for the evolution of advance life forms, the fraction of planets in which a civilisation capable of broadcasting interstellar signals emerges, and the duration for which these kinds of signals are sent out into space. The equation formed the basis of the SETI project. Although the estimates for each factor varies by large amounts, the consensus is that there should be a lot more intelligent aliens in the Milky Way than we have evidence for – which is currently 0. 📺